UNIVERSITY OF PISA

DEPARTMENT OF INFORMATION ENGINEERING

Master's Degree in Robotics and Automation Engineering

Navigation System Project

Navigation System Design for HAPS Drone

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Contents

Chapter 1

Introduction

The objective of this project is to design, simulate, and validate an integrated Inertial Navigation System (INS) aided by a Global Navigation Satellite System (GNSS) and a Attitude and Heading Reference System (AHRS). The system architecture, the kinematic models, and the required sensor suite were strictly dictated by a uniquely assigned project code, which outlines a specific set of algorithmic and structural specifications.

1.1 Project Specifications

The system has been developed according to the assigned code D - E - E - G/a - E - PV, which translates to the following architectural choices:

- **D (Decoupled Estimation):** The navigation system features a decoupled architecture. The attitude estimation is performed separately from the position and velocity estimation, utilizing two distinct filtering instances.

- **E (Ellipsoid Earth Model):** The kinematic mechanization assumes an ellipsoidal Earth model (WGS84). The position state is expressed in geodetic coordinates (Latitude, Longitude, Altitude), requiring the dynamic computation of the meridian and transverse radii of curvature (R_M and R_N).

- **E (Euler Angles Parameterization):** The attitude of the vehicle is parameterized within the state vector using Euler angles (Roll, Pitch, Yaw) rather than quaternions.

- **G/a (Sensor Bias Estimation):** The gyroscope bias is actively estimated within the filter state and fed back to correct the raw measurements. Conversely, the accelerometer bias is not estimated, relying solely on the stochastic noise model.

- **E (Attitude Measurements):** The attitude filter exploits direct measurements of the Euler angles, simulating the output of an Attitude and Heading Reference System (AHRS), to correct the inertial integration.
- **PV (Position and Velocity Measurements):** The navigation filter is aided by direct observations of both 3D position and 3D velocity, representing a typical GNSS receiver output in a loosely coupled configuration.

1.2 Operational Scenario and Trajectory Generation

To properly stress the ellipsoidal kinematics and evaluate the filter's performance over significant Earth-surface displacements, the simulated scenario must involve extended distances and high altitudes. For this reason, the chosen vehicle is a High Altitude Pseudo-Satellite (HAPS), taking the flight characteristics of the Airbus Zephyr as a realistic operational example.

The simulated mission represents a nighttime stratospheric flight with a total duration of 30 minutes (1800 seconds). The vehicle starts at a cruising altitude of 20,000 meters, flying at a highly efficient airspeed of approximately 20 m/s. Due to the massive wingspan and structural constraints typical of HAPS platforms, the generated trajectory is characterized by extremely gentle dynamics, featuring wide turning radii.

Furthermore, to accurately simulate the nighttime power-saving operation, where electric propulsion is minimized in the absence of solar irradiance, a slow and linear altitude loss of 200 meters is distributed over the 30-minute window. This specific scenario highlights the necessity of the WGS84 ellipsoidal model, as a flat-Earth approximation would introduce significant divergence over such extended operational ranges, and provides a challenging dynamic for the vertical channel of the error-state Kalman filter.

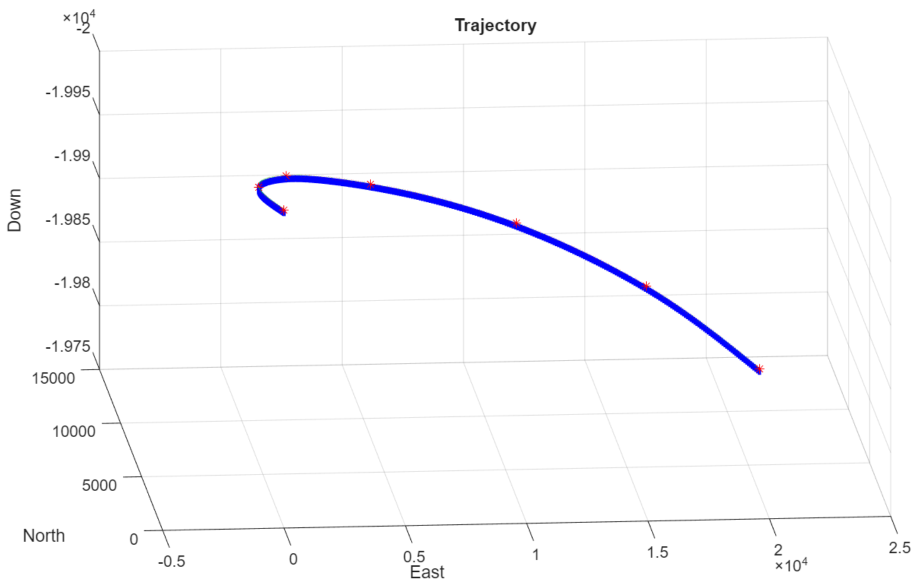

Figure 1.1: Simulated HAPS trajectory in NED frame.

Chapter 2

Inertial Mechanization on the Ellipsoidal Earth

The core of the navigation system is the mechanization algorithm, which integrates the high-frequency measurements provided by the Inertial Measurement Unit (IMU), namely specific forces and angular velocities, to track the vehicle's position, velocity, and attitude. According to the project specifications, the mechanization is formulated over the WGS84 ellipsoidal Earth model, avoiding flat-Earth approximations.

The algorithm is structured into three interconnected modules: position and velocity kinematics, attitude update with transport-rate compensation, and gravity compensation.

2.1 Ellipsoidal Position Kinematics

The vehicle's position is expressed in the geodetic coordinate frame (LLH), comprising latitude, longitude, and altitude. To avoid ambiguity with the roll angle, geodetic latitude is denoted here by φ, while longitude and altitude are denoted by λ and h, respectively. Because of the oblateness of the WGS84 ellipsoid, the Earth's radii of curvature are not constant and must be evaluated as functions of the current latitude.

The prime vertical radius of curvature R_N and the meridian radius of curvature R_M are computed at each time step as

$$R_N = \frac{R_e}{\sqrt{1 - e^2 \sin^2 \varphi}} \tag{2.1}$$

$$R_M = R_N \frac{1 - e^2}{1 - e^2 \sin^2 \varphi} \tag{2.2}$$

where $R_e = 6\,378\,137\,\mathrm{m}$ is the semi-major axis and $e \approx 0.0818$ is the first eccentricity of the ellipsoid.

Using the navigation-frame velocities v_N, v_E, and v_D, the LLH kinematic time derivatives are given by

$$\dot{\varphi} = \frac{v_N}{R_M + h} \tag{2.3}$$

$$\dot{\lambda} = \frac{v_E}{(R_N + h)\cos\varphi} \tag{2.4}$$

$$\dot{h} = -v_D \tag{2.5}$$

These derivatives are numerically integrated to propagate the vehicle's geodetic position.

2.2 Attitude Update and Transport-Rate Compensation

The attitude of the vehicle is parameterized using Euler angles, namely roll ϕ, pitch θ, and yaw ψ, which represent the orientation of the body frame with respect to the local NED frame.

The raw angular velocities measured by the gyroscope, $\boldsymbol{\omega}_{ib}^b$, are referenced to the inertial frame. However, the local NED frame is not stationary: it rotates with respect to the inertial frame because of both the Earth's rotation and the motion of the vehicle over the curved Earth surface. This combined rotation is the transport rate, $\boldsymbol{\omega}_{in}^n$, defined as the sum of the Earth rate $\boldsymbol{\omega}_{ie}^n$ and the craft rate $\boldsymbol{\omega}_{en}^n$:

$$\boldsymbol{\omega}_{ie}^n = \begin{bmatrix} \omega_E \cos\varphi \\ 0 \\ -\omega_E \sin\varphi \end{bmatrix} \tag{2.6}$$

$$\boldsymbol{\omega}_{en}^n = \begin{bmatrix} \dfrac{v_E}{R_N + h} \\ \dfrac{-v_N}{R_M + h} \\ \dfrac{-v_E \tan\varphi}{R_N + h} \end{bmatrix} \tag{2.7}$$

$$\boldsymbol{\omega}_{in}^n = \boldsymbol{\omega}_{ie}^n + \boldsymbol{\omega}_{en}^n \tag{2.8}$$

where $\omega_E \approx 7.2921 \times 10^{-5}\,\mathrm{rad\,s^{-1}}$ is the Earth's rotation rate.

To correctly update the vehicle's attitude, the transport rate must be projected into the body frame via the direction cosine matrix C_n^b and subtracted from the raw gyroscope measurements. The corrected body angular rates are then mapped into Euler-angle derivatives using the standard kinematic Jacobian J:

$$
\begin{bmatrix} \dot{\phi} \\ \dot{\theta} \\ \dot{\psi} \end{bmatrix} = \begin{bmatrix} 1 & \sin\phi\tan\theta & \cos\phi\tan\theta \\ 0 & \cos\phi & -\sin\phi \\ 0 & \dfrac{\sin\phi}{\cos\theta} & \dfrac{\cos\phi}{\cos\theta} \end{bmatrix} \left(\boldsymbol{\omega}_{ib}^{b} - C_n^b \boldsymbol{\omega}_{in}^{n}\right) \tag{2.9}
$$

2.3 Gravity Compensation

The accelerometers measure the specific force $\mathbf{f}^b$, which includes both the kinematic acceleration of the vehicle and the reaction to the Earth's gravitational field. To isolate the true kinematic acceleration needed for the velocity update, the gravity contribution must be subtracted.

Within the WGS84 framework, gravity is not a constant $9.81\,\mathrm{m\,s^{-2}}$. The mechanization adopts the Somigliana closed-form expression to compute the local gravity magnitude $g(\varphi)$ on the ellipsoid surface as a function of latitude:

$$
g(\varphi) = g_{\mathrm{wgs0}} \frac{1 + g_{\mathrm{wgs1}} \sin^2\varphi}{\sqrt{1 - e^2 \sin^2\varphi}} \tag{2.10}
$$

where $g_{\mathrm{wgs0}} \approx 9.7803\,\mathrm{m\,s^{-2}}$ is the gravity at the equator and g_{wgs1} is the gravity formula constant. The resulting local gravity vector is

$$
\mathbf{g}^n = \begin{bmatrix} 0 & 0 & g(\varphi) \end{bmatrix}^T, \tag{2.11}
$$

and is transformed into the body frame and subtracted from the specific-force measurements before the velocity update.

Chapter 3

Sensor Modeling

The accuracy of the navigation filter heavily relies on the realistic modeling of the onboard sensor suite. To simulate a realistic operational environment, the ideal kinematic data was corrupted by introducing stochastic noise and deterministic biases, parameterized based on commercial components. The system integrates a high-frequency (100 Hz) Inertial Measurement Unit (IMU) and a low-frequency (10 Hz) GNSS receiver.

3.1 Inertial Measurement Unit (IMU)

The inertial sensors were modeled using the technical specifications of the 6-axis InvenSense MPU-6050 IMU. Operating at a sampling frequency of $f_s = 100\,\text{Hz}$, the theoretical noise bandwidth (Nyquist limit) is assumed to be $BW = f_s/2 = 50\,\text{Hz}$.

3.1.1 Accelerometer Model

From the "Accelerometer Specifications" section of the datasheet, the following parameters are extracted:

- **Full-Scale Range:** $\pm 2g$
- **Noise Power Spectral Density (ND):** $400\,\mu g/\sqrt{\text{Hz}}$
- **Initial Zero-G Tolerance (Calibration):** $\pm 50\,\text{mg}$ (X and Y axes), $\pm 80\,\text{mg}$ (Z axis)

In accordance with the **G/a** project specification, the accelerometer bias is not included in the filter's state vector and is not estimated. The measurement uncertainty is therefore entirely absorbed by the stochastic noise model.

Converting the noise density into the International System of Units yields:

$$ND_{acc} = 400 \cdot 10^{-6} \cdot 9.81 = 0.003924\,\text{m}\,\text{s}^{-2}/\sqrt{\text{Hz}}. \tag{3.1}$$

Assuming white measurement noise, the theoretical standard deviation of the sensor is obtained by multiplying the noise density by the square root of the bandwidth:

$$\sigma_{\text{th},acc} = ND_{acc} \cdot \sqrt{BW} = 0.003924 \cdot \sqrt{50} \approx 0.0277\,\mathrm{m\,s^{-2}}. \tag{3.2}$$

To ensure the robustness of the Kalman filter, as a conservative approach, and to compensate for the unmodeled bias, high-frequency structural vibrations typical of a large-wingspan HAPS, and mounting misalignments, the nominal standard deviation was increased in the simulator to $\sigma_{acc} = 0.05\,\mathrm{m\,s^{-2}}$. Consequently, the accelerometer noise covariance matrix $R_{acc} \in \mathbb{R}^{3\times3}$ assumes an isotropic diagonal structure:

$$R_{acc} = \mathrm{diag}(\sigma_{acc}^2, \sigma_{acc}^2, \sigma_{acc}^2) = \mathrm{diag}(0.0025, 0.0025, 0.0025)\,(\mathrm{m\,s^{-2}})^2. \tag{3.3}$$

3.1.2 Gyroscope Model

From the "Gyroscope Specifications" section, the following values are obtained:

- **Rate Noise Spectral Density (ND):** $0.005°/\mathrm{s}/\sqrt{\mathrm{Hz}}$
- **Initial Zero-Rate Output (ZRO) Tolerance:** $\pm 20°/\mathrm{s}$

Converting the noise density into radians per second yields:

$$ND_{gyro} = 0.005 \cdot \frac{\pi}{180} \approx 8.727 \cdot 10^{-5}\,\mathrm{rad\,s^{-1}}/\sqrt{\mathrm{Hz}}. \tag{3.4}$$

The theoretical standard deviation is calculated by multiplying the noise density by the square root of the bandwidth:

$$\sigma_{\text{th},gyro} = ND_{gyro} \cdot \sqrt{BW} = 8.727 \cdot 10^{-5} \cdot \sqrt{50} \approx 0.000617\,\mathrm{rad\,s^{-1}}. \tag{3.5}$$

Similar to the accelerometer model, this theoretical standard deviation reflects ideal laboratory conditions. During real flight operations, the sensor is exposed to structural vibrations, temperature-induced drifts, and mechanical noise that increase the actual uncertainty. To adopt a conservative approach and ensure the filter does not over-trust the inertial measurements, the simulated standard deviation was increased to $\sigma_{gyro} = 0.003\,\mathrm{rad\,s^{-1}}$.

Regarding the deterministic error, the datasheet declares an initial ZRO tolerance of $\pm 20°/\mathrm{s}$, which mathematically corresponds to an error of $\pm 0.349\,\mathrm{rad\,s^{-1}}$. However, it is standard practice to assume that the vehicle undergoes a preflight static calibration procedure to estimate and remove the bulk of this gross initial offset. The bias values injected into the simulation, $\mathbf{b}_{gyro} = [0.005, -0.003, 0.006]^T\,\mathrm{rad\,s^{-1}}$, represent the small uncalibrated residual bias. In accordance with the **G/a** specification, the filter dynamically estimates this residual bias during the flight to correct the raw measurements.

3.2 Attitude and Heading Reference System (AHRS)

To estimate the aircraft's attitude, the decoupled system uses direct measurements of the Euler angles. The reference sensor chosen for this simulation is the VectorNav VN-100, a miniature high-performance AHRS module.

From the performance specifications section of the datasheet, the following accuracy values are extracted:

- **Pitch/Roll Accuracy (Dynamic):** 1.0° RMS
- **Magnetic Heading/Yaw Accuracy:** 2.0° RMS
- **Attitude Output Data Rate:** up to 400 Hz

The RMS values provided by the manufacturer were assumed as nominal standard deviations for the measurement noise. To integrate them into the error-state Kalman filter, these values were converted into radians:

$$\sigma_{\text{roll/pitch}} = 1.0 \cdot \frac{\pi}{180} \approx 0.0174\,\text{rad} \tag{3.6}$$

$$\sigma_{\text{yaw}} = 2.0 \cdot \frac{\pi}{180} \approx 0.0349\,\text{rad} \tag{3.7}$$

Since the Euler angles are measured independently, the attitude measurement error covariance matrix assumes a diagonal form:

$$R_{\text{ahrs}} = \text{diag}(\sigma^2_{\text{roll/pitch}}, \sigma^2_{\text{roll/pitch}}, \sigma^2_{\text{yaw}})\,\text{rad}^2. \tag{3.8}$$

3.3 Global Navigation Satellite System (GNSS)

The navigation block requires absolute position and velocity measurements to bound the intrinsic drift of the inertial integration. The receiver modeled in the system is the u-blox NEO-M8N module.

From the commercial component's specifications, the following key parameters are highlighted:

- **Maximum Navigation Update Rate:** 10 Hz
- **Horizontal Position Accuracy (Autonomous):** 2.5 m CEP
- **Velocity Accuracy:** $0.05\,\text{m}\,\text{s}^{-1}$

Since the positional state vector is expressed in WGS84 geodetic coordinates, the planimetric variance must be converted into radians by normalizing it against the Earth's radius R_e:

$$\sigma^2_{\text{pos,horiz}} = \left(\frac{2.5}{R_e}\right)^2 \text{ rad}^2. \tag{3.9}$$

Regarding the vertical axis, GNSS altitude is typically degraded by worse geometric dilution of precision and uncompensable tropospheric delays. For this reason, the altitude standard deviation was conservatively set to 3.5 m in the simulator.

The three-dimensional velocity measurements, derived by the receiver through carrier-phase Doppler shift analysis, maintain the nominal datasheet value, with a standard deviation of $\sigma_{\text{vel}} = 0.05\,\text{m}\,\text{s}^{-1}$.

The GNSS measurement covariance matrix is then built by assembling the position and velocity variances:

$$R_{\text{gps}} = \text{diag}(\sigma^2_{\text{pos,horiz}}, \sigma^2_{\text{pos,horiz}}, 3.5^2, \sigma^2_{\text{vel}}, \sigma^2_{\text{vel}}, \sigma^2_{\text{vel}}). \tag{3.10}$$

Chapter 4

Decoupled INS/GNSS and AHRS Integration via Error-State KF

In accordance with the project specifications, the navigation algorithm implements a Decoupled (D) architecture. Rather than utilizing a single, centralized 15-state filter, the estimation process is strictly divided into two independent instances: an Attitude filter and a position and velocity filter. This architectural choice significantly reduces the computational burden by manipulating smaller covariance matrices (6x6 instead of 15x15).

4.1 The Error-State KF Formulation

The integration between the Inertial Navigation System (INS) and the aiding sensors (AHRS and GNSS) is achieved through an Error-State Kalman Filter (ESKF), also known as an Indirect EKF. Unlike a Direct EKF, which attempts to estimate the total state of the vehicle, the ESKF exclusively estimates the errors accumulated by the inertial mechanization.

This indirect approach provides several fundamental advantages for the system architecture.

First, the filter always operates close to the origin. Because the error states are continuously reset and kept infinitesimally small, the system stays far from parameter singularities and gimbal lock issues, providing a mathematical guarantee that the linear approximation remains valid at all times. This approach also ensures numerical stability, avoiding the floating-point truncation errors that typically occur in Direct EKFs when manipulating large absolute values, such as the Earth's radius or geodetic coordinates.

Second, the linearization process is fast and analytically simple. Since the error state is always small, all second-order products become negligible. This makes the computation of the Jacobians straightforward, completely bypassing the need for heavy numerical differentiation at runtime.

Finally, the error dynamics are inherently slow because all the fast, large-signal dynamics are integrated exactly by the nominal-state mechanization at the IMU frequency of 100 Hz. This feature is particularly crucial for multi-rate architectures. While the AHRS filter can seamlessly apply its corrections at 100 Hz, the slow error growth allows the navigation filter to apply its computationally expensive GNSS corrections at a much lower rate (10 Hz). This ensures that the system effortlessly ingests asynchronous, low-frequency updates without ever sacrificing the high-frequency dynamic tracking capabilities of the vehicle.

4.2 Qualitative Observability Analysis

Before evaluating the filter's performance, it is crucial to perform a qualitative analysis of the state-vector observability. Observability dictates whether the internal states of the system can be inferred from the available external measurements over time. Thanks to the Decoupled (D) architecture and the chosen sensor suite, the observability of both filters is highly robust.

4.2.1 Attitude Filter Observability

The attitude error-state vector contains six variables: the three Euler-angle errors and the three gyroscope biases.

The AHRS provides direct, absolute measurements of roll, pitch, and yaw. Consequently, the three attitude error states are directly and continuously observable.

Regarding the gyroscope biases, their observability is guaranteed by the kinematic coupling of the system. An uncompensated bias in the gyroscope measurements mathematically integrates into a drifting attitude error. Because the AHRS bounds the attitude drift with absolute measurements, the Kalman filter can unambiguously attribute the observed growing attitude discrepancy to the underlying gyroscope biases. Therefore, under normal dynamic conditions, the entire 6-state attitude vector is fully observable.

4.2.2 Navigation Filter Observability

The navigation error-state vector contains six variables: the three 3D geodetic position errors and the three 3D NED velocity errors.

In this architecture, the qualitative observability is trivial. The aiding sensor is a GNSS receiver configured to output both 3D position and 3D velocity measurements. Because every single state in the navigation filter is directly measured by the external sensor, the observation matrix is the identity matrix. There are no hidden or coupled states to infer indirectly.

4.3 Filter Matrices and Asynchronous Multi-Rate Execution

To handle the different output rates of the sensors, namely the AHRS and IMU at 100 Hz and the GNSS at 10 Hz, the filters use an asynchronous logic. The prediction step is executed continuously at the IMU discrete time step $\Delta t = 0.01\,\mathrm{s}$, propagating the error covariance. The correction step is triggered only when a new measurement is available, using an internal counter for the lower-frequency GNSS data. Both filters use the Joseph form for the covariance update to guarantee positive semi-definiteness.

4.3.1 Attitude Error Filter

The state vector for the attitude filter comprises the Euler-angle errors and the gyroscope biases:

$$\delta x_{att} = \begin{bmatrix}\delta\phi, \delta\theta, \delta\psi, \delta b_{gx}, \delta b_{gy}, \delta b_{gz}\end{bmatrix}^T = \begin{bmatrix}\delta\Theta \\ \delta b_g\end{bmatrix} \tag{4.1}$$

To derive the continuous-time error dynamics, we linearize the attitude kinematic equation. The true Euler angles rates are related to the true body angular rates by the kinematic Jacobian J:

$$\dot{\Theta} = J\omega_{nb}^b \tag{4.2}$$

The gyroscope measures the body rates corrupted by the slowly varying bias b_g and the wide-band noise w_g. By applying the perturbation method ($\tilde{x} = x+\delta x$), ignoring the Earth and transport rates errors due to the decoupled assumption, the differential equation for the Euler-angle error $\delta\Theta$ is obtained as:

$$\delta\dot{\Theta} = -J\delta b_g - Jw_g \tag{4.3}$$

The gyroscope bias is modeled as a random walk process driven by white noise w_{bg}:

$$\delta\dot{b}_g = w_{bg} \tag{4.4}$$

Arranging these differential equations in matrix form ($\delta\dot{x}_{att} = F_{att}\delta x_{att} + G_{att}w_{att}$), we obtain the continuous-time error dynamics matrix F_{att} and the noise mapping matrix G_{att}:

$$F_{att} = \begin{bmatrix}0_{3\times3} & -J \\ 0_{3\times3} & 0_{3\times3}\end{bmatrix} \tag{4.5}$$

$$G_{att} = \begin{bmatrix} -J & 0_{3\times3} \\ 0_{3\times3} & I_{3\times3} \end{bmatrix} \tag{4.6}$$

where J is evaluated using the current mechanized attitude:

$$J = \begin{bmatrix} 1 & \sin\phi\tan\theta & \cos\phi\tan\theta \\ 0 & \cos\phi & -\sin\phi \\ 0 & \frac{\sin\phi}{\cos\theta} & \frac{\cos\phi}{\cos\theta} \end{bmatrix} \tag{4.7}$$

The discrete-time state transition matrix F_k and noise matrix D_k are approximated as

$$F_k \approx I_{6\times6} + F_{att}\Delta t \tag{4.8}$$

$$D_k \approx G_{att}\Delta t. \tag{4.9}$$

During the correction step, the innovation is computed against the AHRS measurements y_{ahrs}. The observation matrix H_{att} isolates the first three states:

$$H_{att} = [I_{3\times3}, 0_{3\times3}]. \tag{4.10}$$

4.3.2 Navigation Error Filter

The state vector for the navigation filter comprises the geodetic position errors and the NED velocity errors:

$$\delta x_{nav} = \left[\delta\varphi, \delta\lambda, \delta h, \delta v_N, \delta v_E, \delta v_D\right]^T = \begin{bmatrix} \delta P \\ \delta V^n \end{bmatrix} \tag{4.11}$$

For the position states, the kinematics are governed by the radii of curvature matrix T_r. The perturbation of $\dot{P} = T_r V^n$ yields the position error dynamics:

$$\delta\dot{P} = T_r \delta V^n \tag{4.12}$$

Regarding the velocity error dynamics, the decoupled architecture simplifies the standard mechanization equations. By neglecting the Coriolis and gravity error coupling (which are negligible for the gentle dynamics of the HAPS scenario), the velocity error is primarily driven by the uncompensated accelerometer noise w_a (since accelerometer bias is not estimated according to the **G/a** specification):

$$\delta\dot{V}^n = w_a \tag{4.13}$$

Arranging these into the continuous-time state-space formulation ($\delta\dot{x}_{nav} = F_{nav}\delta x_{nav} + G_{nav}w_{nav}$), the resulting system matrices are:

$$F_{nav} = \begin{bmatrix} 0_{3\times3} & T_r \\ 0_{3\times3} & 0_{3\times3} \end{bmatrix} \tag{4.14}$$

$$T_r = \begin{bmatrix} \frac{1}{R_M+h} & 0 & 0 \\ 0 & \frac{1}{(R_N+h)\cos\varphi} & 0 \\ 0 & 0 & -1 \end{bmatrix} \tag{4.15}$$

$$G_{nav} = \begin{bmatrix} 0_{3\times3} \\ I_{3\times3} \end{bmatrix} \tag{4.16}$$

The GNSS provides direct measurements of all six navigation states. Consequently, the measurement matrix H_{nav} is an identity matrix:

$$H_{nav} = I_{6\times6}. \tag{4.17}$$

Because the GNSS updates at 10 Hz, the correction step is executed every 10 iterations of the prediction loop. Upon correction, the estimated errors are extracted and added back to the inertial mechanization, effectively resetting the filter state to zero and closing the indirect Kalman loop.

4.3.3 Filter Tuning: Covariance Matrices Q and R

The performance of the decoupled architecture relies heavily on the definition of the process-noise covariance Q and the measurement-noise covariance R.

- **Measurement Noise (R):** The matrices R_{ahrs} and R_{gps} strictly contain the measurement variances derived analytically from the commercial sensor datasheets, as detailed in Chapter 3. These values represent the statistical confidence in the external observations.

- **Process Noise (Q):** Unlike R, the process-noise matrix must encapsulate unmodeled vehicle dynamics, structural vibrations, and integration discretization errors. Consequently, Q was tuned empirically through a heuristic trial-and-error approach during simulation testing. A nominal base variance of 0.01 was selected for the primary diagonal elements, corresponding to accelerometer and gyroscope noise, to provide a balanced filter response, preventing an over-reliance on the pure kinematic prediction while avoiding the amplification of high-frequency measurement noise. Furthermore, for the attitude filter, the elements of Q corresponding to the bias random walk were scaled down by a factor of 10^{-3} to correctly reflect the slow-varying, highly stable nature of the in-run sensor instability.

Chapter 5

Simulation Results and Performance Analysis

The final testing phase evaluates the performance of the decoupled error-state Kalman filter during the simulated 30-minute stratospheric flight. The closed-loop filtered estimation is directly compared against the pure inertial mechanization, which runs open-loop and integrates the raw, corrupted IMU data without any external aiding.

5.1 Attitude and Bias Estimation Performance

The first phase of the analysis focuses on the performance of the decoupled attitude filter. Running the pure inertial mechanization open-loop for the 30-minute stratospheric flight results in significant, fluctuating attitude errors. Rather than diverging to infinity immediately, the pure INS estimation heavily oscillates and gradually drifts away from the true kinematics. This behavior is primarily driven by the uncompensated gyroscope biases and noise being integrated continuously over time.

The introduction of the Error-State Kalman Filter, aided by the AHRS measurements at 100 Hz, effectively bounds these fluctuations. As shown in Figure 5.2, the ESKF maintains the attitude estimation tightly coupled to the ground-truth kinematics, completely suppressing the drift and reducing the error to zero-mean noise typical of a well-tuned filter.

To quantitatively evaluate the filter's performance, the statistical metrics are compared in Table 5.1. The table highlights the massive improvement introduced by the ESKF. While the pure INS accumulates large absolute errors and an RMS yaw error of over 8.18 rad during the flight, the closed-loop ESKF drastically reduces these values to the order of 10^{-3} rad. Furthermore, the ESKF achieves a mean error of exactly zero across all axes, proving that the estimation is perfectly unbiased and operates well within the nominal accuracy of the selected AHRS module.

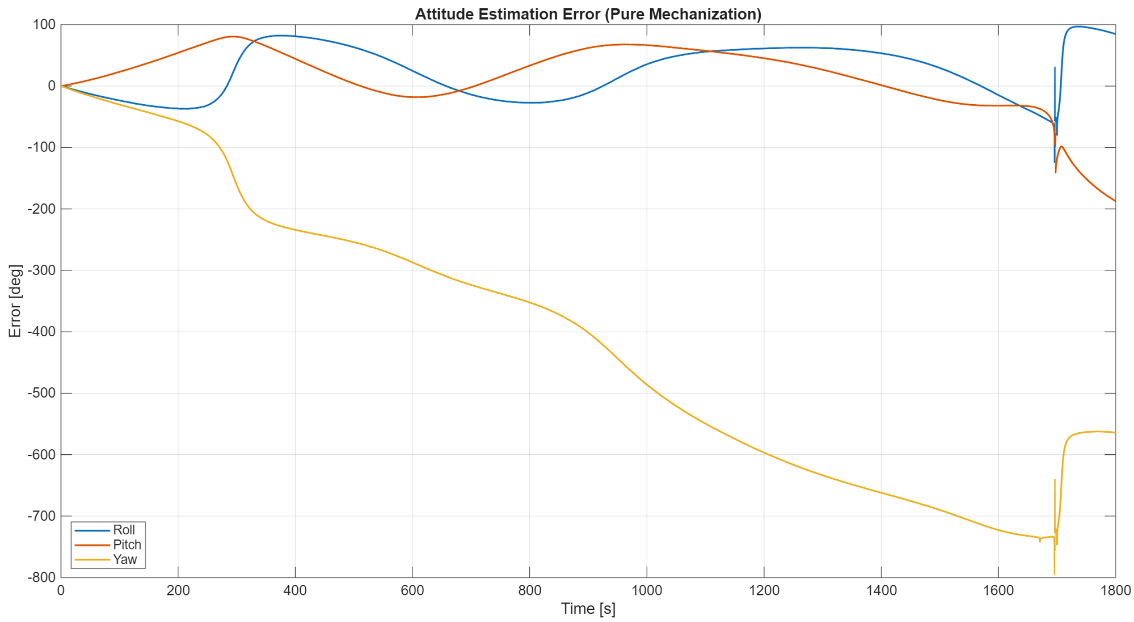

Figure 5.1: Time evolution of the attitude estimation error resulting from pure mechanization, with no filter applied.

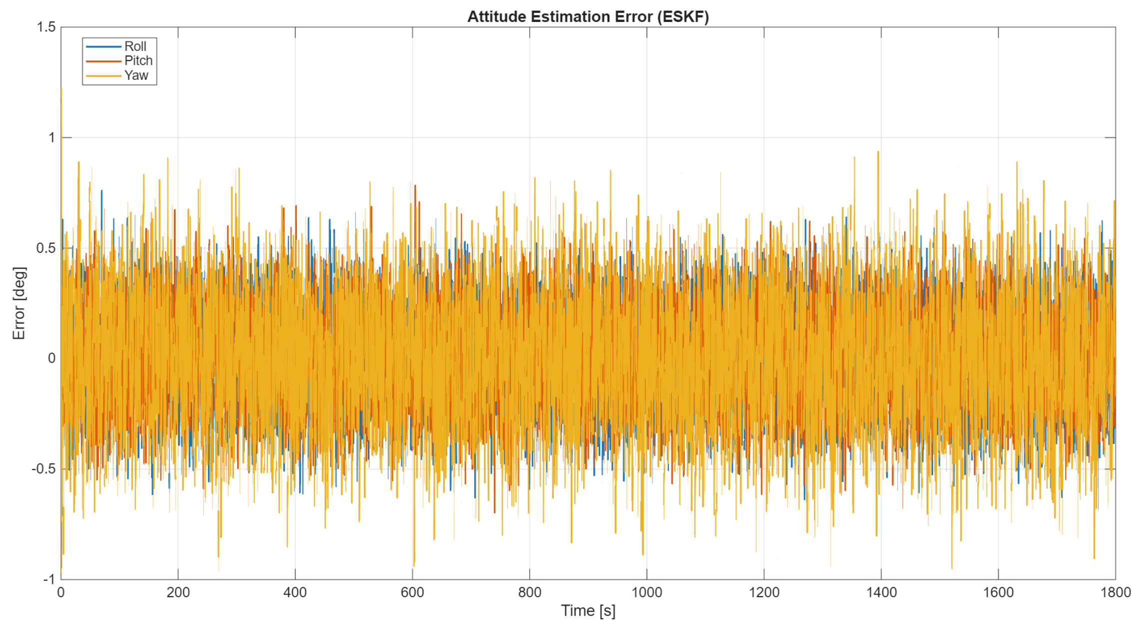

Figure 5.2: Time evolution of the attitude estimation error resulting from pure mechanization, with the ESKF filter applied.

Table 5.1: Statistical metrics comparison for attitude estimation (Pure Mechanization vs. ESKF).

Configuration	Statistic	Roll [rad]	Pitch [rad]	Yaw [rad]
Pure Mech.	Max Abs Error	2.1643	3.2689	13.8673
Pure Mech.	Mean Error	0.4080	0.2439	-7.1589
Pure Mech.	RMS Error	0.8369	0.9440	8.1845
ESKF	Max Abs Error	0.0165	0.0136	0.0213
ESKF	Mean Error	0.0000	0.0000	0.0000
ESKF	RMS Error	0.0030	0.0030	0.0042

A critical aspect of the filter's success is its ability to dynamically estimate the deterministic sensor errors. The in-run gyroscope biases, which were initialized at zero in the filter state, rapidly converge to the true constant values injected into the simulation environment. This validates the system's compliance with the **G/a** specification, ensuring that the mechanization step is fed with continuously corrected angular rates.

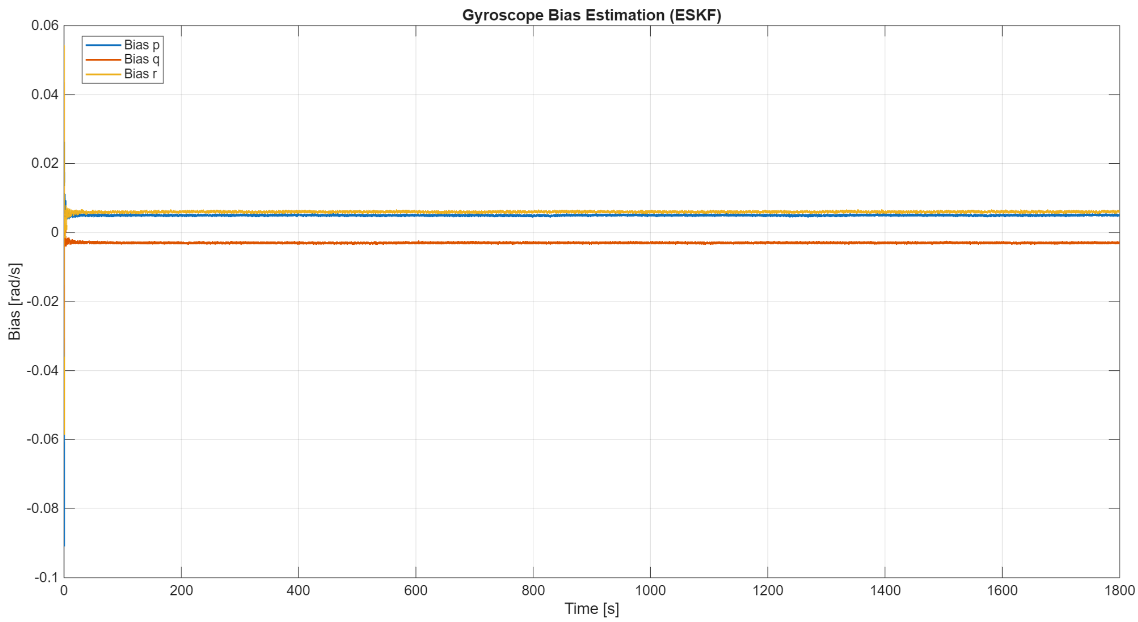

Figure 5.3: Time evolution of the gyroscope biases estimated by the ESKF.

The statistical evaluation of the bias estimation is summarized in Table 5.2. Expressing the rates in the body frame (p, q, r), the results show that once the initial transient has settled, the mean values of the estimated biases perfectly align with the nominal uncalibrated offsets defined in the sensor model (0.005, -0.003, and 0.006 rad/s respectively). This excellent tracking confirms the robust observability and optimal tuning of the attitude filter.

Table 5.2: Statistical metrics for the estimated gyroscope biases (ESKF).

Statistic	Bias p [rad/s]	Bias q [rad/s]	Bias r [rad/s]
Max Abs Value	0.0909	0.0526	0.0583
Mean Value	0.0050	-0.0030	0.0060
RMS Value	0.0050	0.0030	0.0060

To further visualize the filter's noise attenuation capabilities, the actual state trajectories are compared against the noisy measurements and the filtered estimates. Figures 5.4, 5.5, and 5.6 display a highly zoomed-in time window of the attitude tracking. The raw AHRS measurements are visibly corrupted by the high-frequency stochastic noise injected into the simulation. In contrast, the ESKF solution seamlessly tracks the ground-truth dynamics, effectively smoothing the measurement noise without introducing any noticeable phase lag.

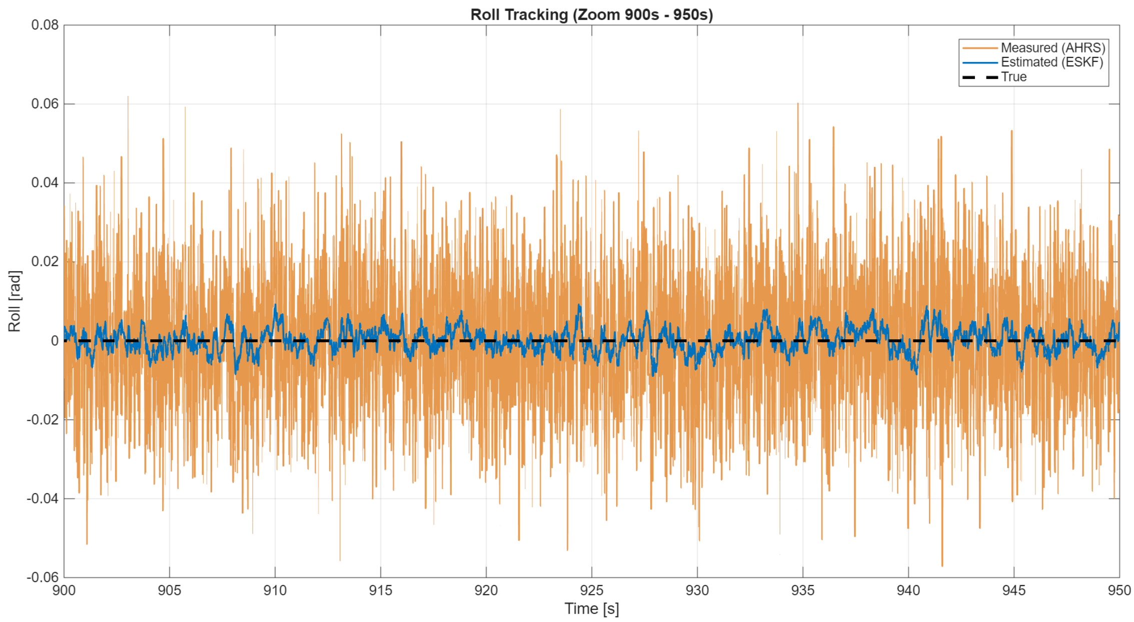

Figure 5.4: Roll trajectory tracking (zoomed): Comparison between ground-truth, noisy AHRS measurement, and the smoothed ESKF estimation.

To explicitly quantify this noise attenuation, the measurement errors (pure sensor noise) are plotted directly against the filter's estimation errors in Figures 5.7, 5.8, and 5.9. The high-frequency stochastic noise introduced by the AHRS is visibly bounded and reduced by the ESKF. These plots demonstrate the filter's capability to effectively filter out the measurement scatter, yielding a

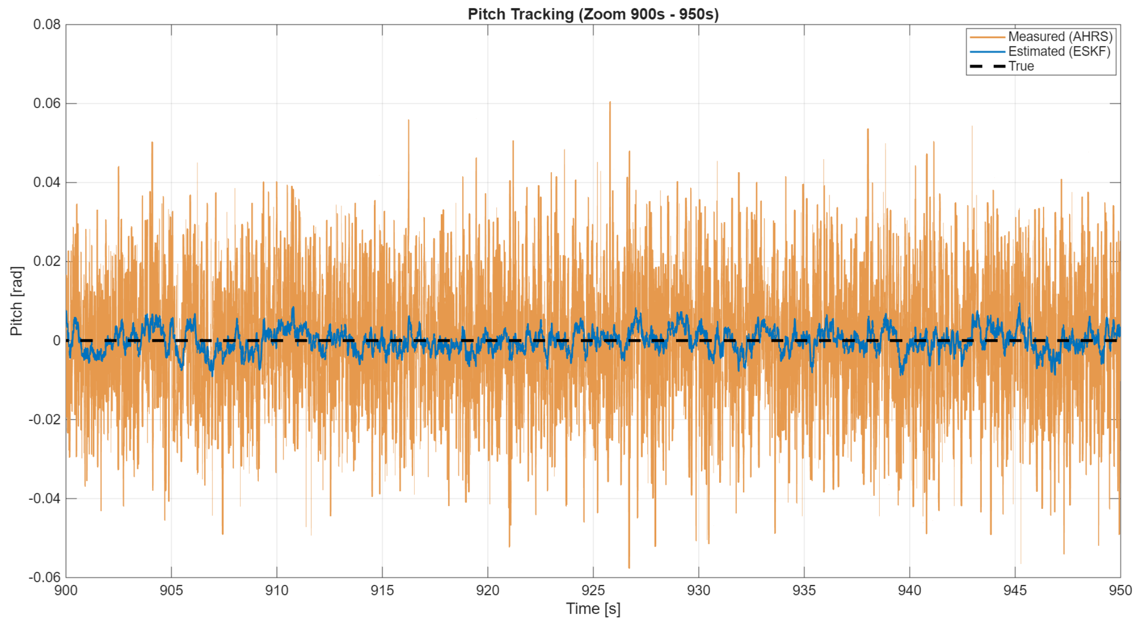

Figure 5.5: Pitch trajectory tracking (zoomed): Comparison between ground-truth, noisy AHRS measurement, and the smoothed ESKF estimation.

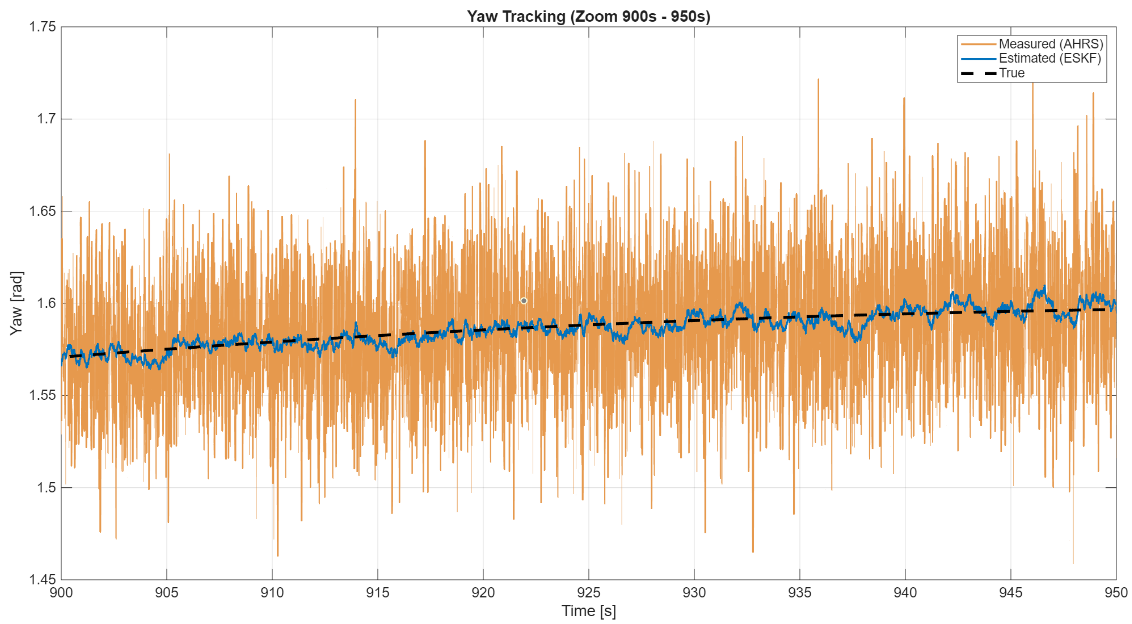

Figure 5.6: Yaw trajectory tracking (zoomed): Comparison between ground-truth, noisy AHRS measurement, and the smoothed ESKF estimation.

clean attitude solution that mitigates the intrinsic inaccuracies of the single sensors.

Figure 5.7: Roll error comparison (zoomed): AHRS measurement error vs. ESKF estimation error.

5.2 Position and Velocity Estimation Performance

The second phase of the analysis evaluates the navigation filter. The 3D position and velocity estimation represents the most challenging aspect of the HAPS scenario. Due to the double integration of uncompensated accelerometer noise, the open-loop INS position and velocity suffer from catastrophic drift over the 1800-second flight.

As detailed in Table 5.3 and Figure 5.10, the pure mechanization completely loses track of the vehicle. By the end of the simulation, it accumulates altitude errors in the order of thousands of kilometers (RMS of $\sim 3 \times 10^6$ m) and horizontal geodetic errors exceeding 1 radian, rendering the pure INS solution totally unusable.

The GNSS-aided ESKF, operating asynchronously to process the 10 Hz measurements, drastically corrects this behavior. Figures 5.11 and 5.12 illustrate the stable, bounded estimation errors provided by the closed-loop architecture.

The filter successfully collapses the horizontal position error down to the order of 10^{-7} radians (equivalent to a few meters physically), achieving a

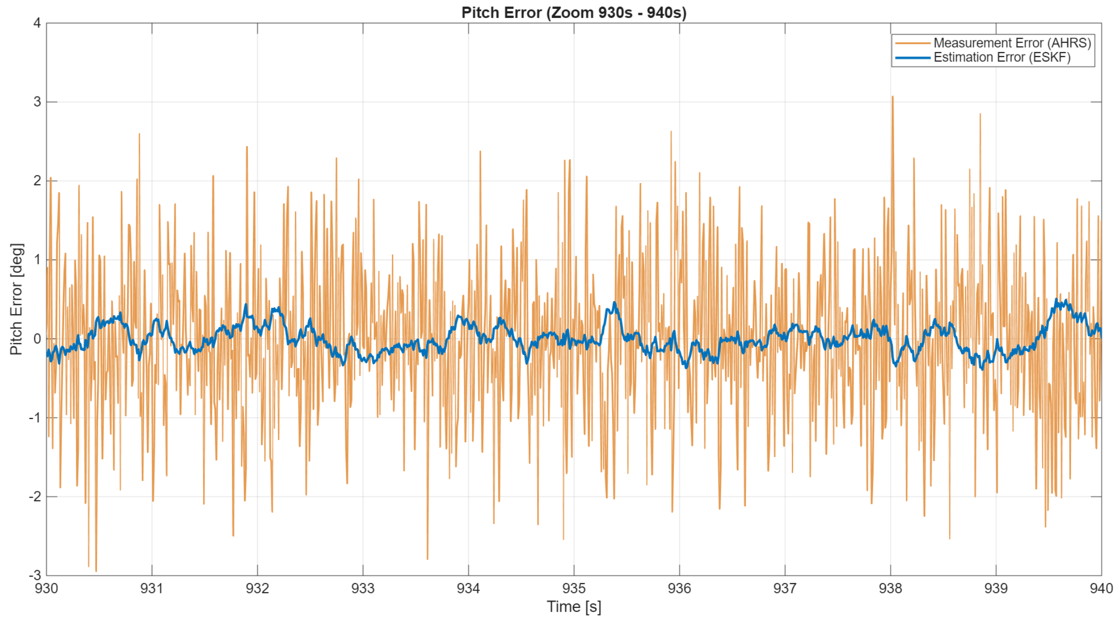

Figure 5.8: Pitch error comparison (zoomed): AHRS measurement error vs. ESKF estimation error.

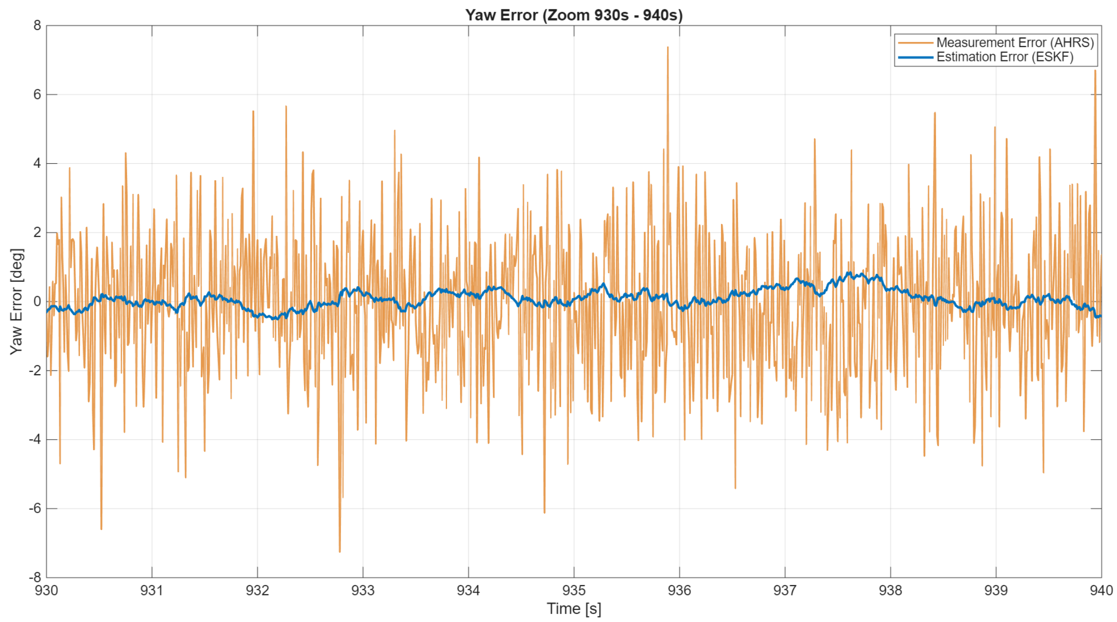

Figure 5.9: Yaw error comparison (zoomed): AHRS measurement error vs. ESKF estimation error.

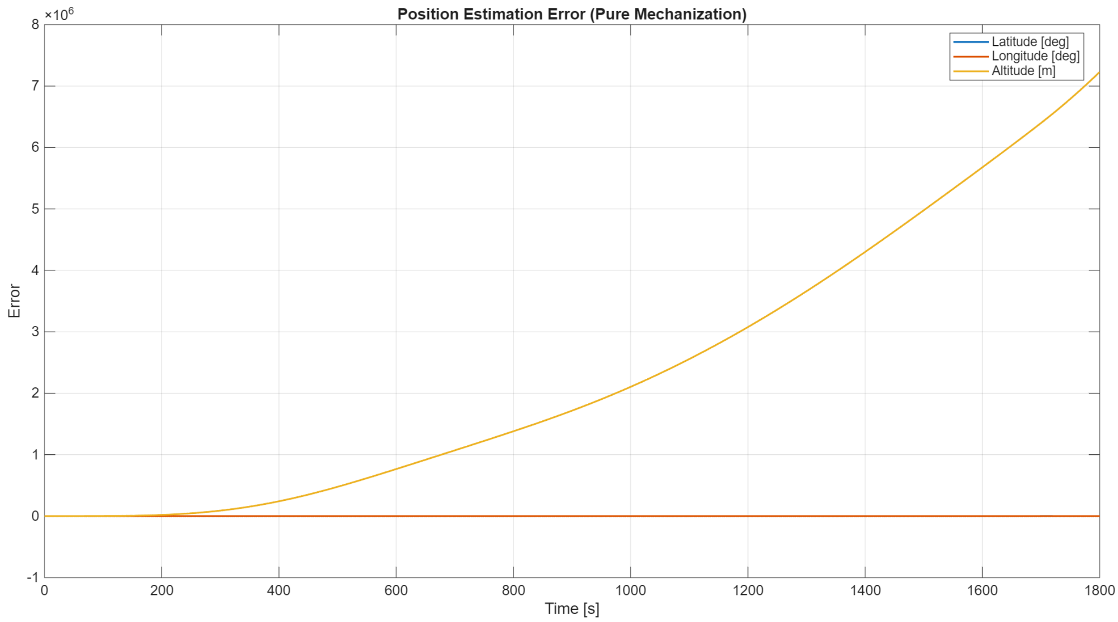

Figure 5.10: Time evolution of the position estimation error resulting from pure mechanization, with no filter applied.

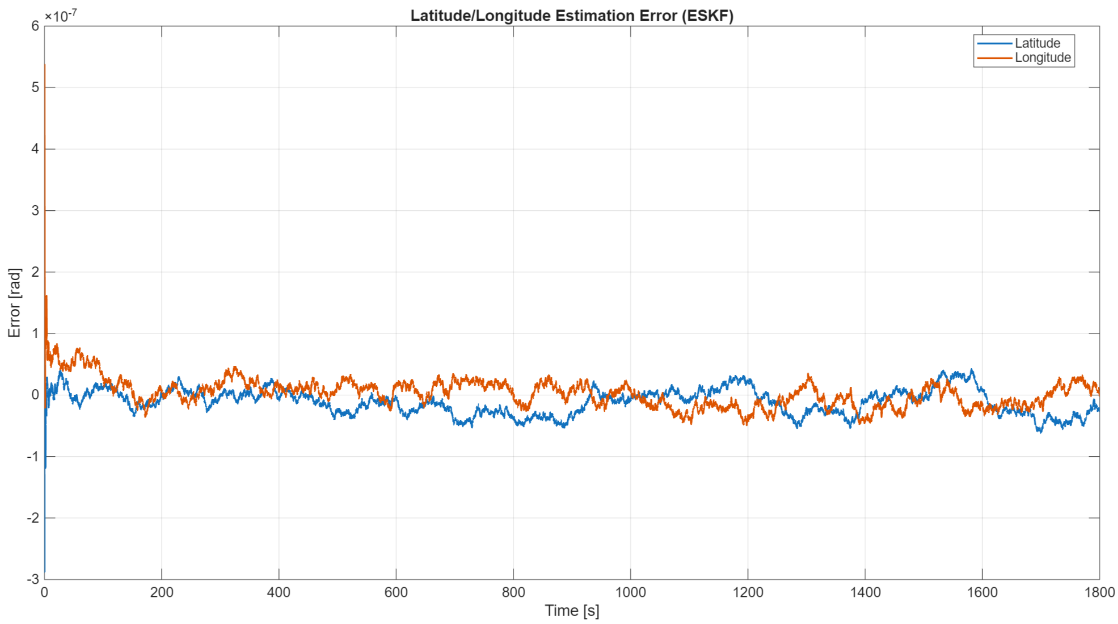

Figure 5.11: Time evolution of the latitude/longitude estimation error resulting from pure mechanization, with ESKF applied.

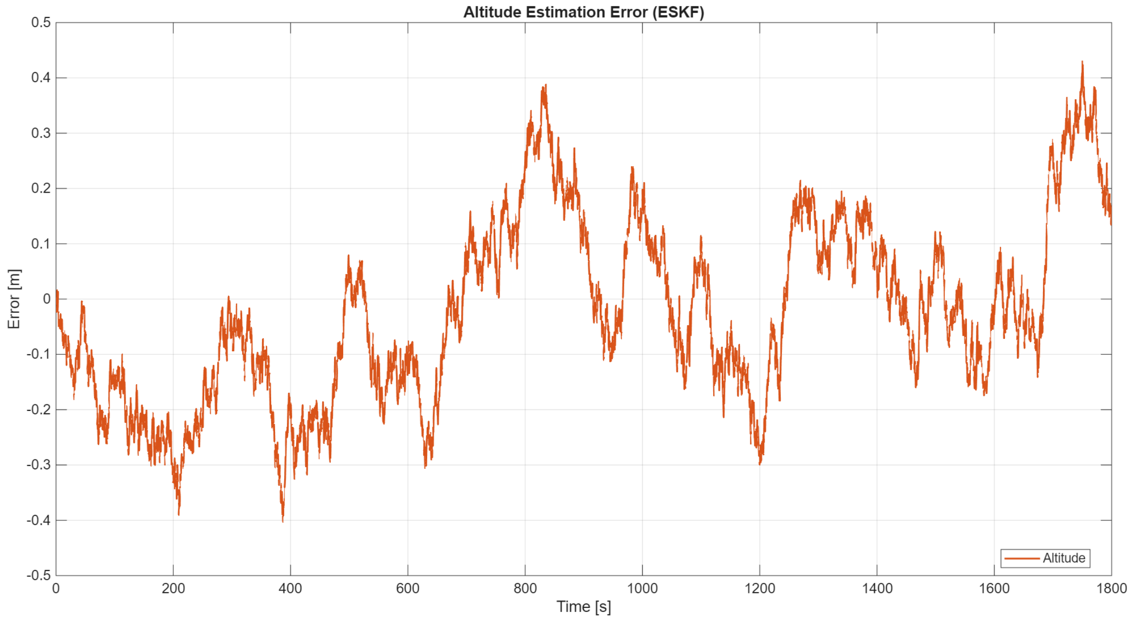

Figure 5.12: Time evolution of the altitude estimation error resulting from pure mechanization, with ESKF applied.

strictly zero-mean error. Furthermore, the vertical channel, notoriously the most susceptible to drift, remains perfectly bounded. The ESKF achieves a steady-state altitude RMS error of just 0.1658 m, exceptionally outperforming the conservative 3.5 m standard deviation expected from the GNSS receiver.

Table 5.3: Statistical metrics comparison for LLH estimation (Pure Mechanization vs. ESKF).

Configuration	Statistic	Latitude [rad]	Longitude [rad]	Altitude [m]
Pure Mech.	Max Abs Error	4.3110	1.2696	$\sim 7 \times 10^6$
Pure Mech.	Mean Error	0.6111	-0.4004	$\sim 2 \times 10^6$
Pure Mech.	RMS Error	1.0283	0.5003	$\sim 3 \times 10^6$
ESKF	Max Abs Error	$\sim 2 \times 10^{-7}$	$\sim 5 \times 10^{-7}$	0.4301
ESKF	Mean Error	0.0000	0.0000	-0.0240
ESKF	RMS Error	0.0000	0.0000	0.1658

A similar massive divergence is observed in the unfiltered velocity kinematics, where accumulated errors reach thousands of meters per second due to sensor noise integration.

The asynchronous correction logic efficiently handles the 10 Hz update rate, preventing the high-frequency IMU noise from degrading the low-frequency stability provided by the GNSS. As detailed in Table 5.4, the filtered velocity

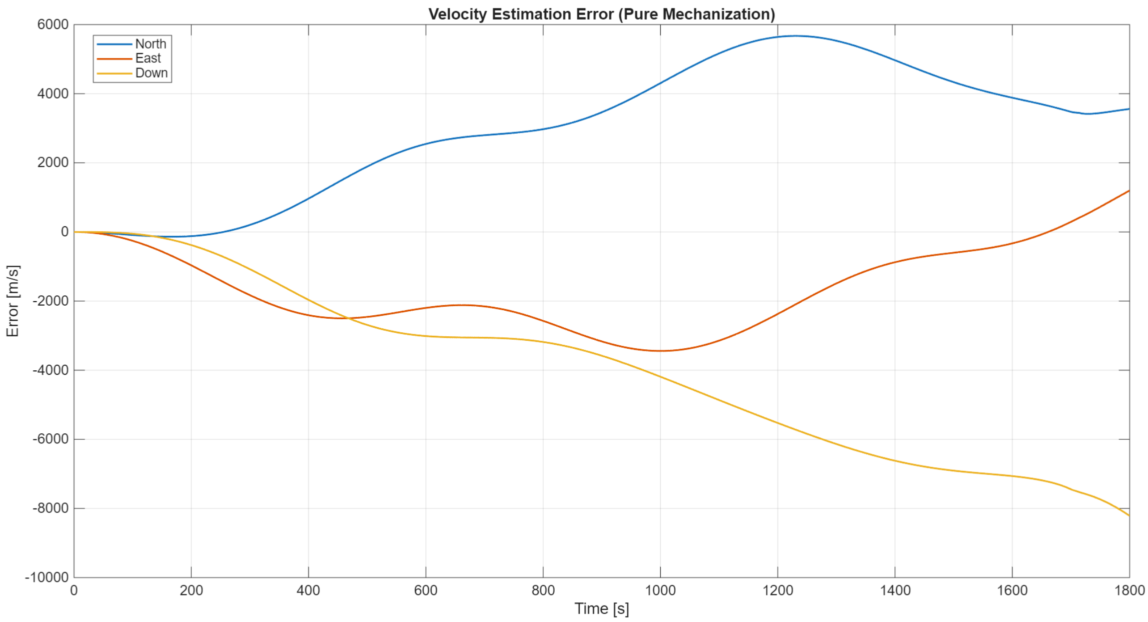

Figure 5.13: Time evolution of the velocity estimation error resulting from pure mechanization, with no filter applied.

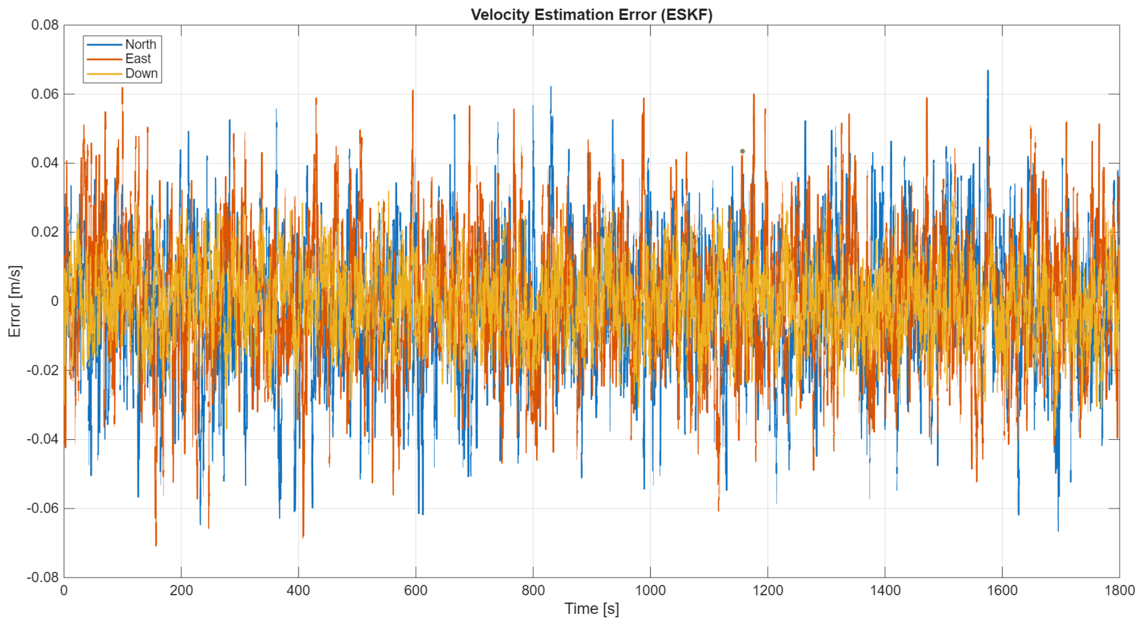

Figure 5.14: Time evolution of the velocity estimation error resulting from pure mechanization, with ESKF applied.

errors maintain an RMS below 0.018 m/s across all axes, which is substantially better than the 0.05 m/s nominal accuracy of the GNSS module. The mean velocity errors are strongly centered around zero, demonstrating the robustness of the decoupled architecture even during a prolonged stratospheric flight.

Table 5.4: Statistical metrics comparison for velocity estimation (Pure Mechanization vs. ESKF).

Configuration	Statistic	v_N [m/s]	v_E [m/s]	v_D [m/s]
Pure Mech.	Max Abs Error	~ 5672	~ 3443	~ 8214
Pure Mech.	Mean Error	~ 2984	~ 1636	-3986
Pure Mech.	RMS Error	~ 3530	~ 2024	~ 4666
ESKF	Max Abs Error	0.0668	0.0707	0.0387
ESKF	Mean Error	-0.0017	0.0006	-0.0005
ESKF	RMS Error	0.0180	0.0179	0.0100

The noise rejection performance is equally evident when analyzing the kinematic trajectories. Combining GPS and INS has the primary goal of limiting the estimation drift while simultaneously improving the absolute navigation solution. Figures 5.15, 5.16, and 5.17 show a highly zoomed-in perspective of the velocity components. The raw GNSS measurements exhibit discrete, low-rate (10 Hz) noisy updates. The closed-loop ESKF optimally fuses this information with the high-rate (100 Hz) inertial integration, yielding a smooth velocity estimate that closely follows the true trajectory, filtering out the GNSS scatter.

Similarly, the position tracking performance is depicted in Figures 5.18, 5.19, and 5.20. Despite the significant variance characterizing the standalone GNSS positional fixes, the integrated system effectively rejects the measurement noise. The estimated latitude, longitude, and altitude remain tightly anchored to the ground-truth path, visually confirming the superior accuracy and reliability of the integrated navigation system over uncoupled sensors.

Finally, the robustness of the decoupled architecture is validated by directly comparing the raw GNSS measurement errors against the ESKF estimation errors. Figures 5.21, 5.22, and 5.23 illustrate the velocity error attenuation, where the discrete measurement scatter is significantly damped by the filtering process.

Similarly, Figures 5.24, 5.25, and 5.26 demonstrate the position error reduction. The ESKF effectively bounds the GNSS positional inaccuracies, providing an estimation error that is continuously smoothed and kept well within the raw sensor's uncertainty limits, further proving the success of the data fusion algorithm.

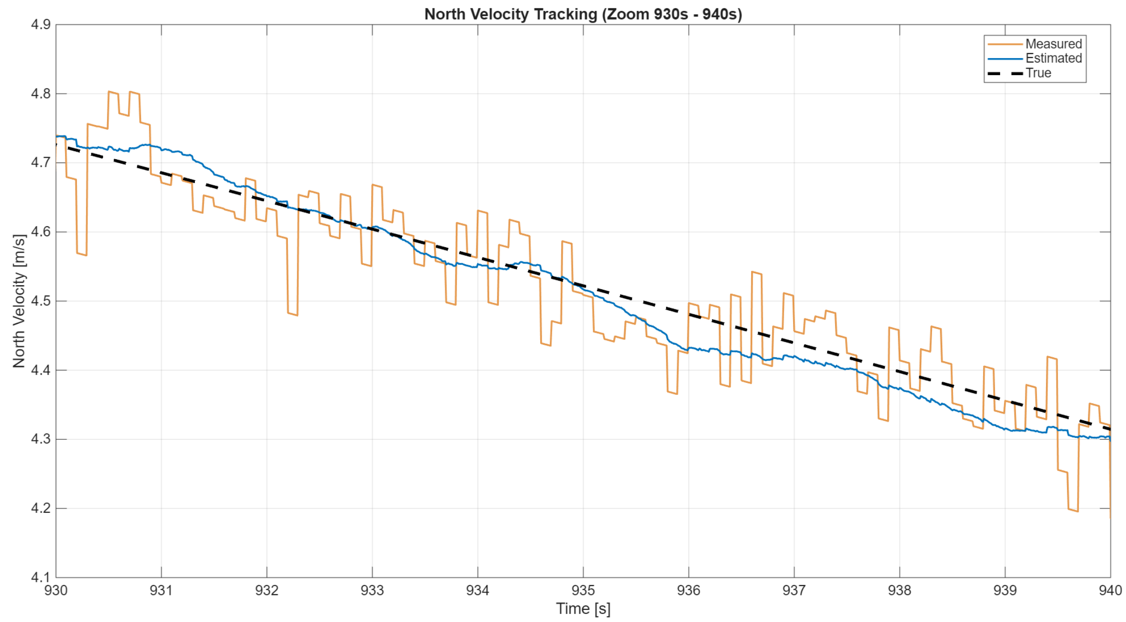

Figure 5.15: North velocity trajectory tracking (zoomed): Comparison between ground-truth, noisy GNSS measurement, and the ESKF estimation.

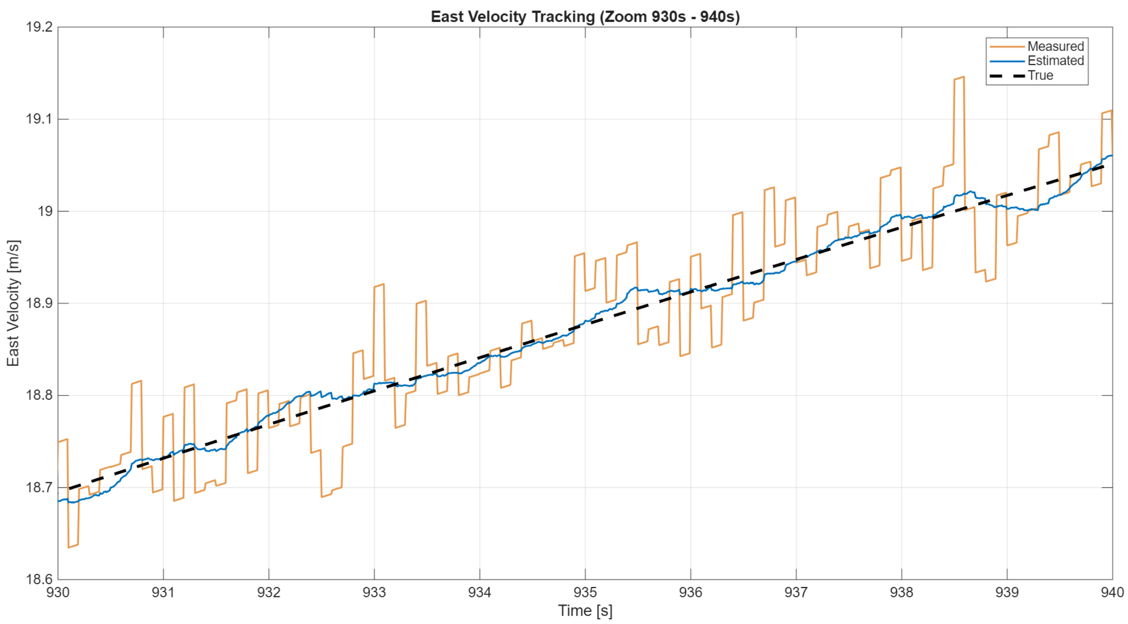

Figure 5.16: East velocity trajectory tracking (zoomed): Comparison between ground-truth, noisy GNSS measurement, and the ESKF estimation.

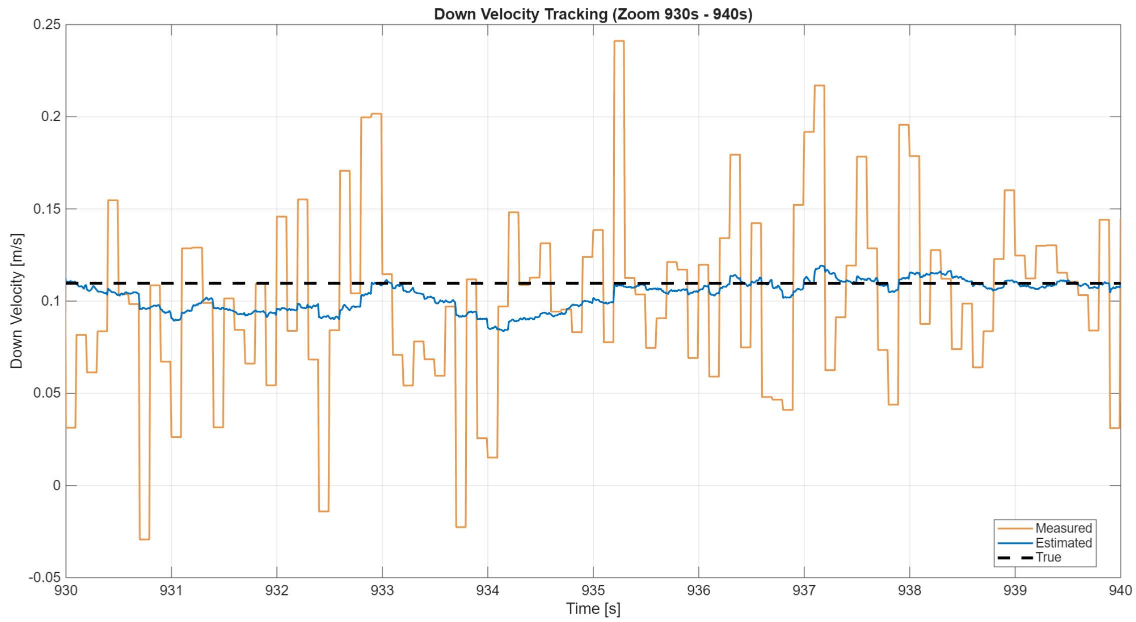

Figure 5.17: Down velocity trajectory tracking (zoomed): Comparison between ground-truth, noisy GNSS measurement, and the ESKF estimation.

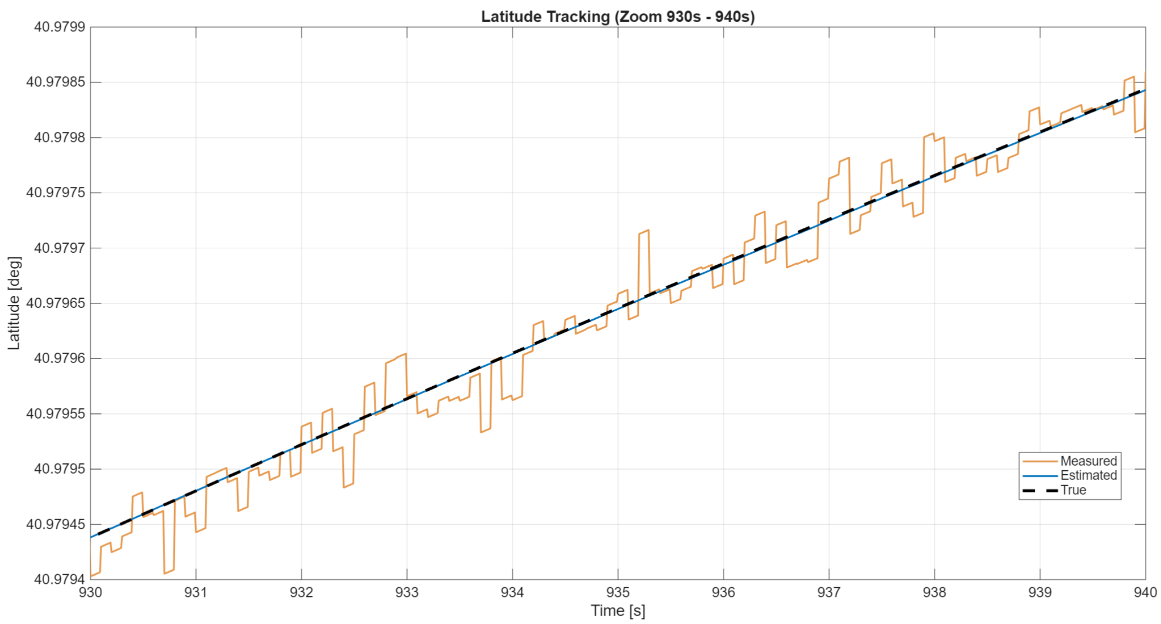

Figure 5.18: Latitude trajectory tracking (zoomed): Comparison between ground-truth, noisy GNSS measurement, and the ESKF estimation.

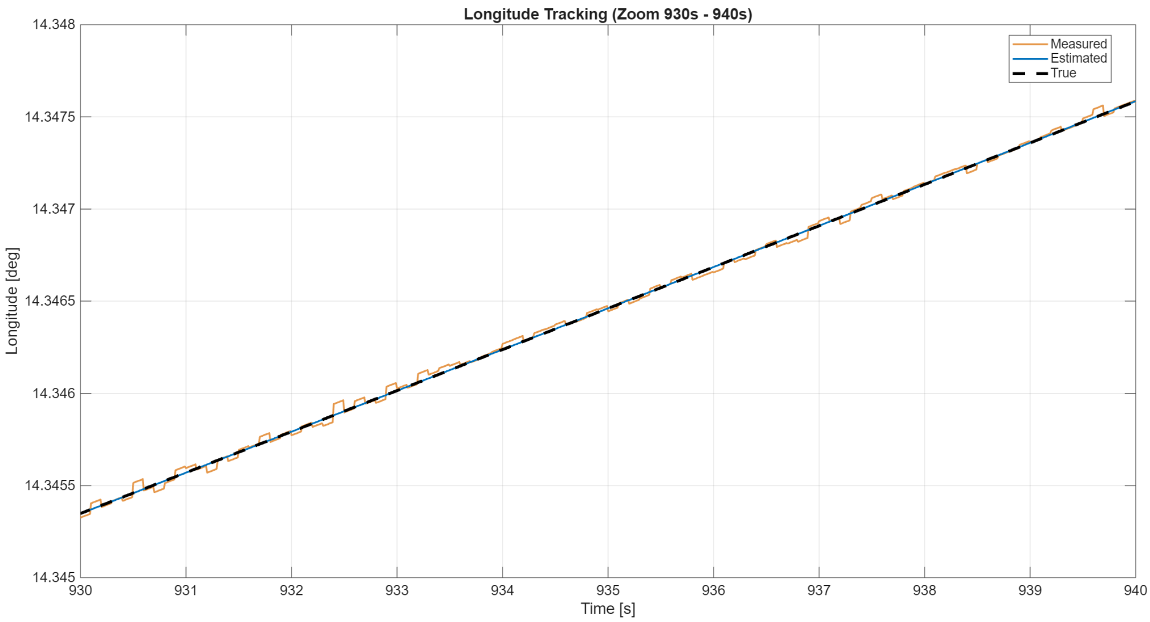

Figure 5.19: Longitude trajectory tracking (zoomed): Comparison between ground-truth, noisy GNSS measurement, and the ESKF estimation.

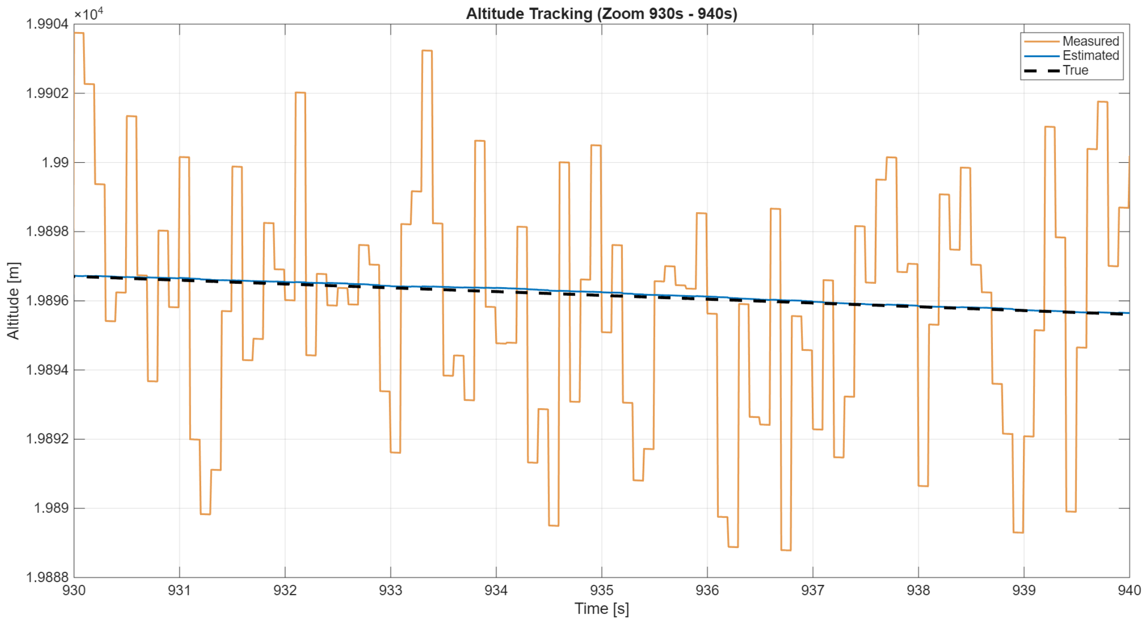

Figure 5.20: Altitude trajectory tracking (zoomed): Comparison between ground-truth, noisy GNSS measurement, and the ESKF estimation.

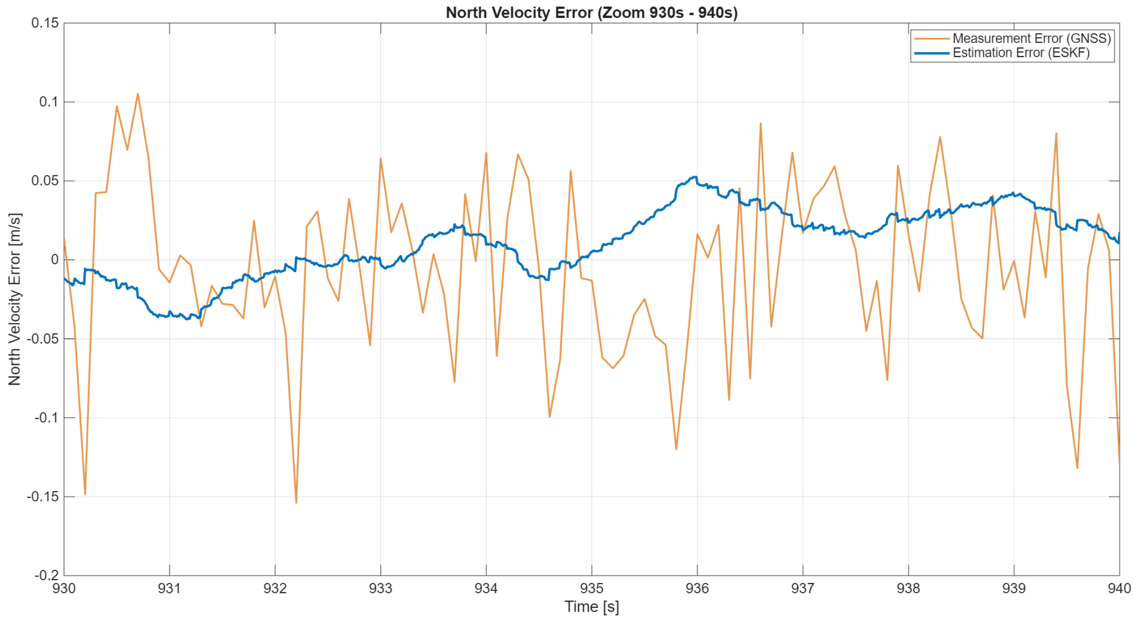

Figure 5.21: North velocity error comparison (zoomed): GNSS measurement error vs. ESKF estimation error.

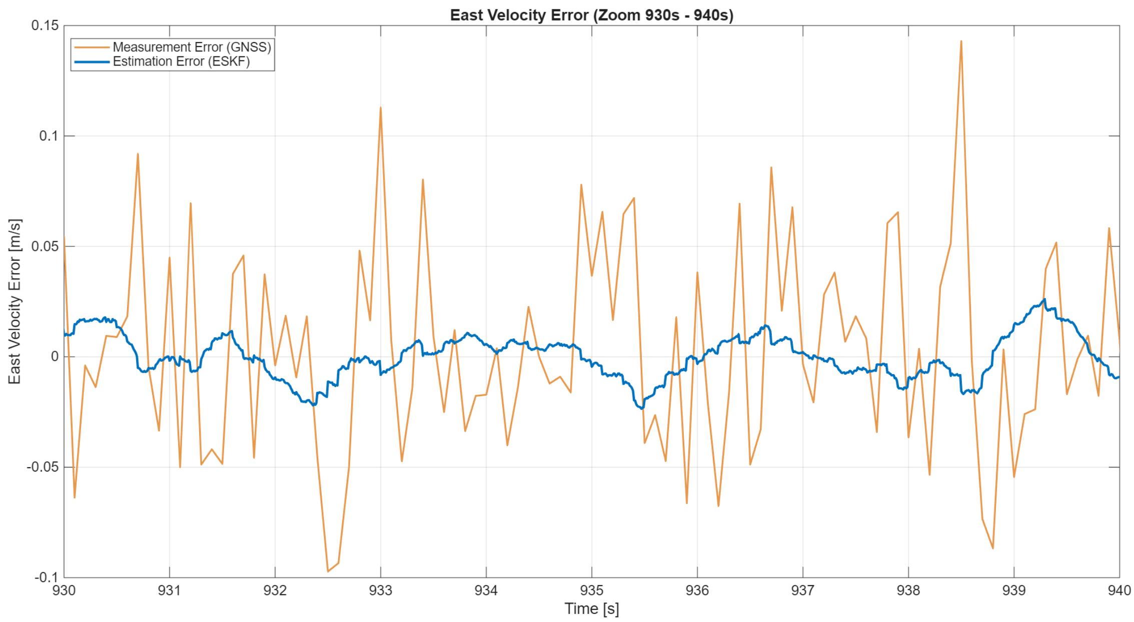

Figure 5.22: East velocity error comparison (zoomed): GNSS measurement error vs. ESKF estimation error.

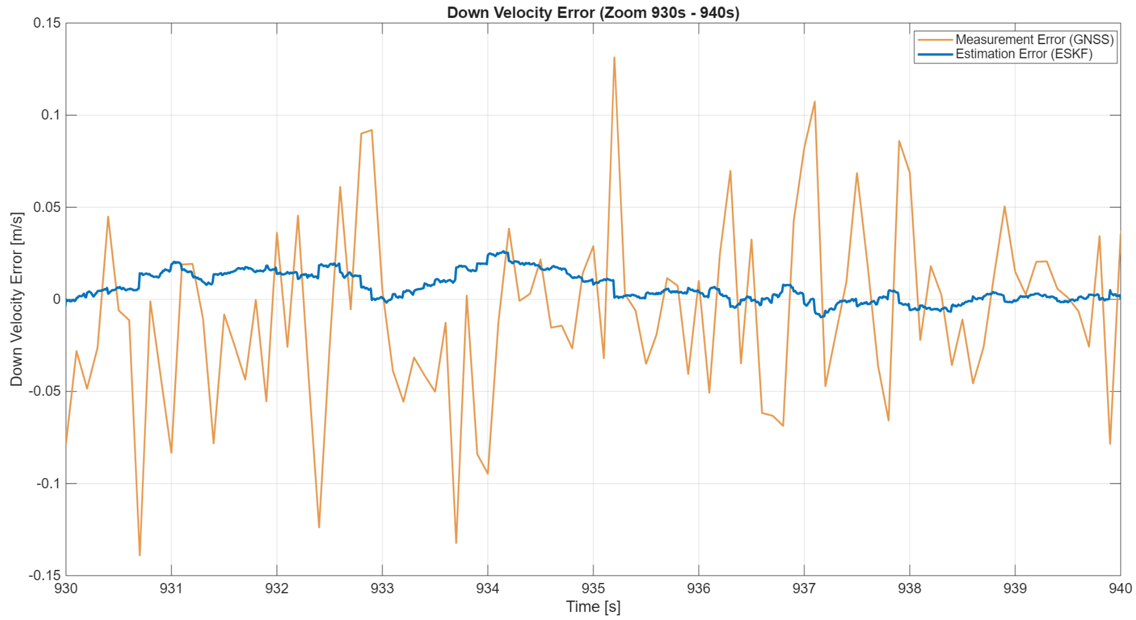

Figure 5.23: Down velocity error comparison (zoomed): GNSS measurement error vs. ESKF estimation error.

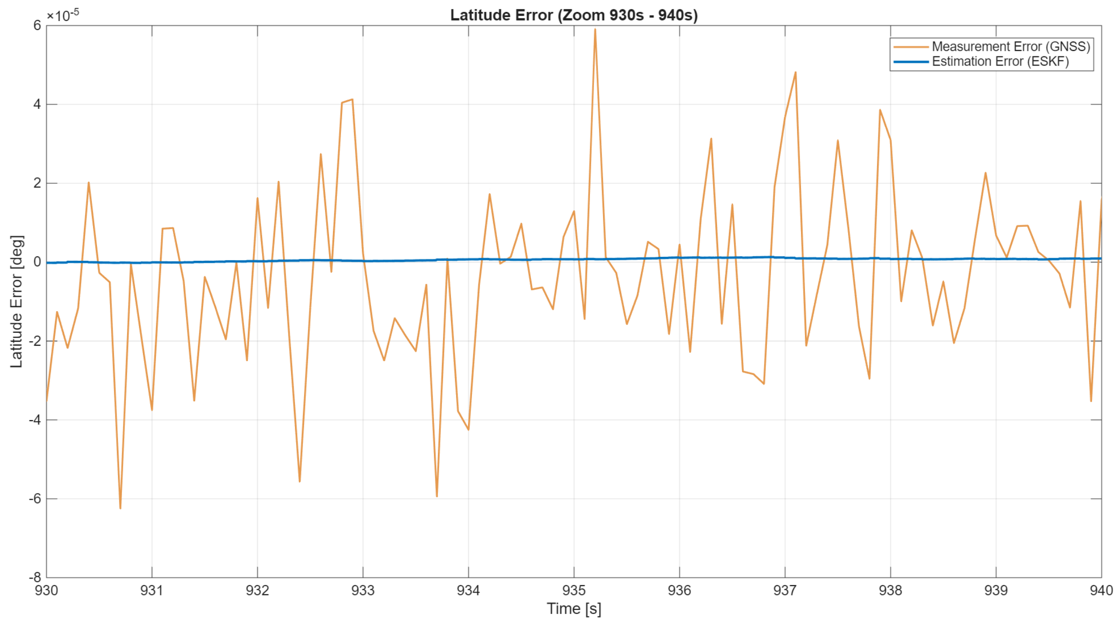

Figure 5.24: Latitude error comparison (zoomed): GNSS measurement error vs. ESKF estimation error.

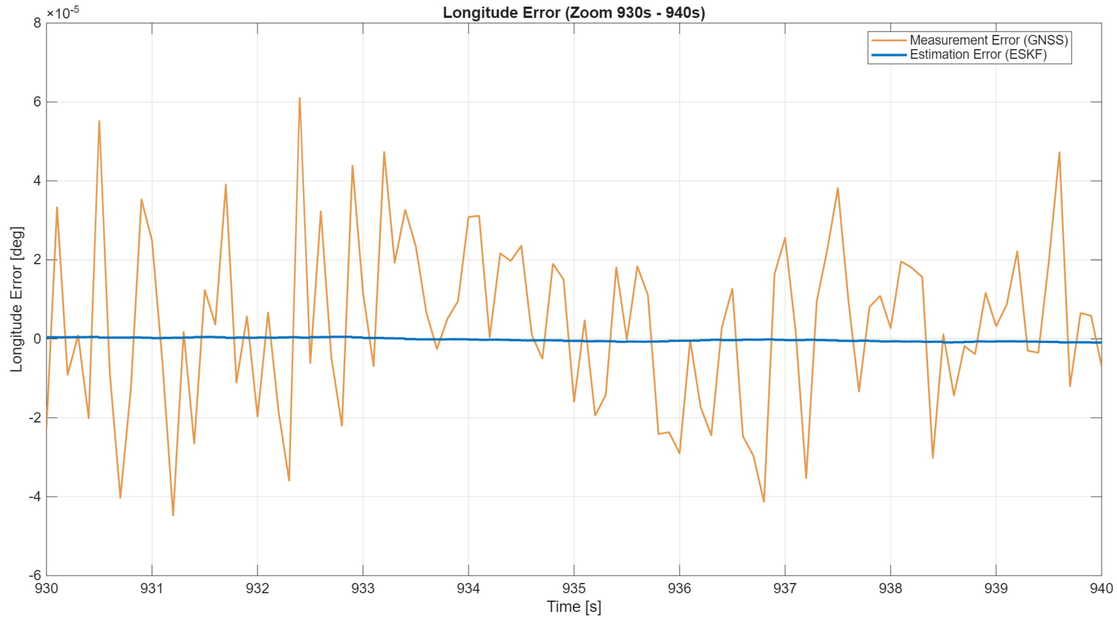

Figure 5.25: Longitude error comparison (zoomed): GNSS measurement error vs. ESKF estimation error.

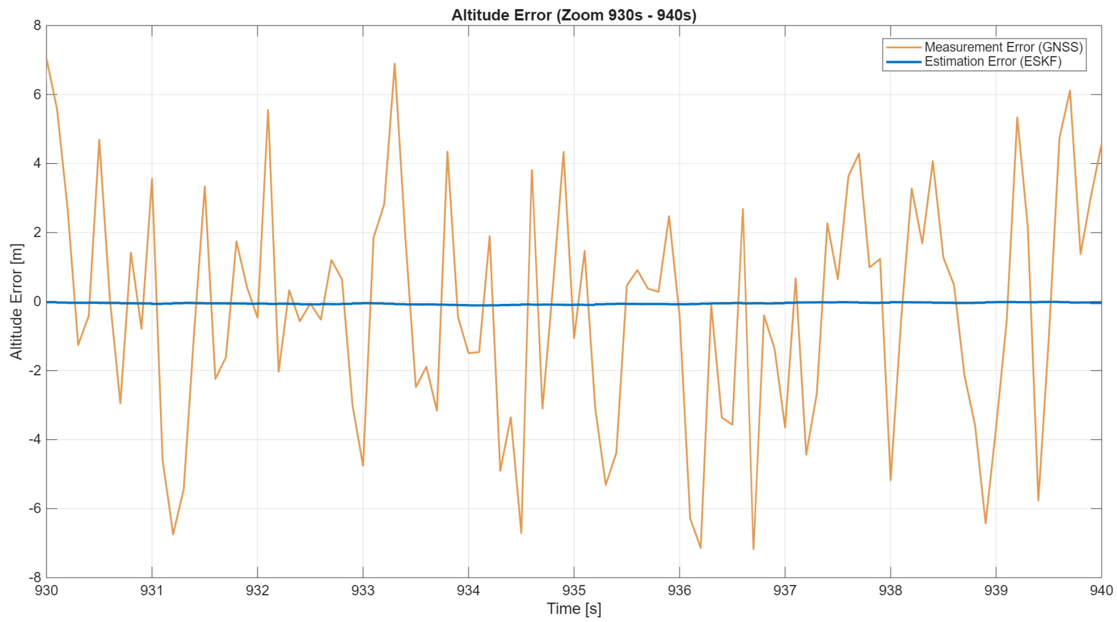

Figure 5.26: Altitude error comparison (zoomed): GNSS measurement error vs. ESKF estimation error.

Chapter 6

Conclusions

By implementing a Decoupled Error-State Kalman Filter, the system effectively bounded the catastrophic drift inherent to pure inertial mechanization. The asynchronous multi-rate integration of high-frequency AHRS data and low-frequency GNSS measurements proved highly efficient. The attitude filter successfully isolated and compensated for the deterministic in-run gyroscope biases, while the navigation filter maintained steady-state position and velocity errors well within the nominal accuracy bounds of the selected commercial sensor suite.

Ultimately, the proposed indirect filtering architecture demonstrated excellent numerical stability and robust tracking capabilities, confirming its suitability and reliability for long-endurance stratospheric flight operations.